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EDA dashboard

ML Models



 Air Quality · M3

UCI Dataset · Italy · 2004–2005

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Mechanical Engineer → Data Analyst

M3 · 2026



Air Quality Analysis · ML Engine

UCI Air Quality Dataset · Italian Road-Level
Sensor Station · March 2004 – April 2005
End-to-end EDA · Time Series · Statistical
Testing · A/B Test · Machine Learning



Dataset at a Glance



Records



Features



Period



Missing...



Sensors

9,... 15 M... 13... 5



About This Project



**Business
/ Research
Problem**

Can we predict urban
air pollution levels from
low-cost metal-oxide
sensors — and identify
the key drivers of high
CO concentrations in a
real Italian city?



Analysis Roadmap

1

Data Overview

2

Univariate Analysis

3

Bivariate Analysis

4

Correlation Analysis

5

Feature Engineering

6

Missing Values &
Imputation

Dataset — UCI Air Quality

Hourly sensor readings from a multisensor device deployed in a polluted area of an Italian city. Contains ground-truth (GT) measurements from a certified reference analyzer alongside 5 metal-oxide sensor responses.

Key Data Challenge

-200 is the sensor null marker (not real data) — replaced with NaN.
NMHC(GT) has 90% missing → dropped from analysis.
CO(GT), NO_x, NO₂ have ~19% missing → median imputation.

7 Pollutant Time Trends 

8 Weather vs Pollution 

9 A/B Test (Rush Hour) 

10 Time Series Decomposition

11 ADF & Statistical Tests

12 Multicollinearity VIF 

13 Insights & Recommendations

Machine Learning Targets

Regression Target

CO(GT) — Carbon Monoxide ground truth (mg/m³)
Predict hourly CO concentration from

Classification Target

high_pollution — Binary (CO > median → 1, else → 0)
Predict whether the current hour is a high-

sensor readings + weather
+ time features.

pollution event. ~50/50
balanced.

M3 · Air Quality ML Engine · UCI Dataset · 2026 · Built with Python · Streamlit ·
Scikit-learn · Plotly

